

future, more work will take place remotely, with fewer people on the shop floor.

While terms such as AI, Gen AI and ML are a part of the lexicon now, the challenge is to make them relevant to the manufacturing process, say experts. “Technology will not replace human excellence... but competence levels vary across the workforce,” says Banerjee, adding that AI’s role is to remove that variability.

However, the western template of Industry 4.0 (replacing humans with automation) will not work in India since livelihoods stand to be affected, say experts. Instead, one

will emerge when global supply chains also relocate to India to feed into Indian manufacturing,” he says.

Mathur says the government’s ambition to increase manufacturing’s share to 25% of the GDP is an unmissable opportunity. “At Siemens, we believe digitalisation, with its inherent capability to sustainably drive efficiency, productivity, scalability and flexibility, will act as an enabler and accelerator for manufacturing,” he says, citing the example of how the company transformed its 50-year-old low-voltage switchgear factory in Kalwa (an industrial area outside Mumbai) into a “world-class digitalised” one.

practices, developing a skilled workforce, and robust policies as some of the areas that need to be addressed.

“Gen AI tells me the propensity of a risk/accident and that is done in seconds. Likewise, if I am interviewing someone for a position, AI will tell me the person is not authentic or is reading off the internet because of its high-level search capability,” says Tata Steel’s Banerjee.

SHIFTING BASE

Disruption in the sector is moving at a frenetic pace. But, according to Subhabrata Sengupta, Partner at Avalon Consulting, the pandemic slowed the trend of increased globalisation. “Nearshoring in manufacturing is becoming more prevalent and it is Turkey and Eastern Europe for EU or Mexico for the US,” he says, adding that the question now is whether advances in 3D printing technology could lead to hyper customisation taking place through neighbourhood manufacturing shops.

But it will take time for global manufacturing to shift to India. Ram of Cummins thinks it will be influenced by a combination of capital investments, innovation and disruptive technologies. Each of them has a critical role to play—capital is needed to implement new technologies and up quality, while innovation is what propels product development and process optimisation. AI and other disruptive technologies such as IoT, blockchain and additive manufacturing can revolutionise manufacturing and open up new markets. “These disruptions lead to major shifts in industry dynamics, introduce new leaders and displace incumbents,” says Ram.

Each sector comes with its own unique requirements. Amol Nagar, Executive Director (Global Supply Chain) at GE Aerospace, refers to

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must bring in effectiveness with empathy; instead of knocking off a part of the workforce, one can upskill them, say experts.

FROM THE PRESENT TO THE FUTURE

Sunil Mathur, MD & CEO of enterprise, infra and energy solutions provider Siemens Ltd, says the government’s investments over time in reducing logistics costs, improving infrastructure, and consistent policy measures have helped put India on the path to transforming into a global manufacturing hub. “The tipping point

Experts, however, highlight that a lot still needs to be done on the path to 2047. Ashwath Ram, MD of manufacturing major Cummins Group in India, says if India aspires to be among the Top 3 nations in terms of GDP, it is necessary to both expand and strengthen manufacturing. “A thriving manufacturing sector will ensure self-reliance, boost exports and see job creation,” he says. Ram lists adoption of smart manufacturing technologies (Industry 4.0 practices including AI, ML and automation to enhance efficiency, reduce costs and raise quality), more R&D investments, sustainability and environmental